

MS650A - 2S24

Exercícios da aula de 21/10

(1) Classifique quanto ao tipo as EDP a seguir:

- (a) $\sin^2 x u_{xx} + \sin 2x u_{xy} + \cos^2 x u_{yy} = 0$,
- (b) $u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 0$,
- (c) $(xy - 1)u_{xx} + (x + 2y)u_{xy} + u_{yy} + xy^2u = 0$,
- (d) $yu_{xx} + 2xu_{xy} + yu_{yy} = 0$.

(2) Encontre a forma canônica das seguintes EDPs:

- (a) $u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 0$,
- (b) $u_{xx} + 2u_{xy} + u_{yy} + 5u_x + 3u_y + u = 0$,
- (c) $u_{xx} - 6u_{xy} + 12u_{yy} + 4u_x - u = 0$.

(3) Encontre a forma canônica e integre o resultado obtido para escrever a forma geral das soluções das EDPs abaixo:

- (a) $u_{xx} + yu_{yy} + (1/2)u_y = 0 \quad (y < 0)$,
- (b) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 0$,
- (c) $u_{xx} + (10/3)u_{xy} + u_{yy} = 0$,
- (d) $3u_{xx} + 5u_{xy} + 2u_{yy} + u_x + u_y = -1/3$.

(4) Resolva os seguintes problemas de Cauchy, onde $u = u(x, t)$:

- | | |
|---|--|
| (a) $\begin{cases} u_{tt} - c^2u_{xx} = 0, \\ u(x, 0) = 0, \\ u_t(x, 0) = x. \end{cases}$ | (b) $\begin{cases} u_{tt} - c^2u_{xx} = 0, \\ u(x, 0) = \sin x, \\ u_t(x, 0) = 1. \end{cases}$ |
| (c) $\begin{cases} u_{tt} - c^2u_{xx} = 0, \\ u(x, 0) = x, \\ u_t(x, 0) = x^2. \end{cases}$ | (d) $\begin{cases} u_{tt} - c^2u_{xx} = 0, \\ u(x, 0) = \cos x, \\ u_t(x, 0) = c \sin x. \end{cases}$ |
| (e) $\begin{cases} u_{tt} - c^2u_{xx} = x, \\ u(x, 0) = 0, \\ u_t(x, 0) = 1. \end{cases}$ | (f) $\begin{cases} u_{tt} - c^2u_{xx} = e^x, \\ u(x, 0) = 0, \\ u_t(x, 0) = 1. \end{cases}$ |
| (g) $\begin{cases} u_{tt} - c^2u_{xx} = xe^{-t}, \\ u(x, 0) = 1, \\ u_t(x, 0) = 0. \end{cases}$ | (h) $\begin{cases} u_{xx} + 2u_{xt} - 3u_{tt} = 0, \\ u(x, 0) = \sin x, \\ u_t(x, 0) = 1. \end{cases}$ |

1)

a) $\sin^2 x u_{xx} + \sin 2x u_{xy} + \cos^2 x u_{yy} = 0 \Rightarrow A = \sin^2 x, B = \sin 2x, C = \cos^2 x$

$$B^2 - 4AC = (\sin 2x)^2 - 4\sin^2 x \cos^2 x = (\sin 2x)^2 - (2\sin x \cos x)(2\sin x \cos x) = (\sin 2x)^2 - \sin 2x \sin 2x = 0$$

$B^2 - 4AC = 0$, portanto a E.D.P. é do tipo parabólica.

b) $u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 0 \Rightarrow A=1, B=4, C=3$

$$B^2 - 4AC = 16 - 4 \cdot 1 \cdot 3 = 4 > 0, \text{ como } B^2 - 4AC, \text{ a E.D.P. é do tipo hiperbólica.}$$

c) $(xy-1)u_{xx} + (x+2y)u_{xy} + u_{yy} + xy^2u = 0 \Rightarrow A=xy-1, B=x+2y, C=1$

$B^2 - 4AC = (x+2y)^2 - 4(xy-1) = x^2 + 4xy + 4y^2 - 4xy + 4 = x^2 + 4y^2 + 4 > 0, \forall x, y \in \mathbb{R}. B^2 - 4AC > 0$, então a E.D.P. é do tipo hiperbólica.

d) $y u_{xx} + 2x u_{xy} + y u_{yy} = 0 \Rightarrow A=y, B=2x, C=y$

$$B^2 - 4AC = 4x^2 - 4y^2 = 4(x-y)(x+y), \text{ temos que analisar cada caso separadamente.}$$

$$4(x-y)(x+y) > 0 \Leftrightarrow (x-y > 0 \wedge x+y > 0) \vee (x-y < 0 \wedge x+y < 0) \Leftrightarrow (x > y \wedge x > -y) \vee (x < y \wedge x < -y) \\ \Leftrightarrow (x > y) \vee (x < -y)$$

Logo $B^2 - 4AC > 0 \Leftrightarrow x > y \text{ ou } x < -y$, sendo assim, a E.D.P. do tipo hiperbólico.

$$4(x-y)(x+y) = 0 \Leftrightarrow x-y=0 \vee x+y=0 \Leftrightarrow x=y \vee x=-y, \text{ A E.D.P. é parabólica se } x=y \text{ ou } x=-y.$$

$$4(x-y)(x+y) < 0 \Leftrightarrow (x-y < 0 \wedge x+y > 0) \vee (x-y > 0 \wedge x+y < 0) \Leftrightarrow (x < y \wedge x > -y) \vee (x > y \wedge x < -y) \\ \Leftrightarrow (-y < x < y) \vee (y < |x|)$$

A E.D.P. é elíptica se $-y < x < y$ ou $y < |x|$.

2)

$$a) u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 0 \Rightarrow A=1, B=4, C=3, D=3, E=-1, F=2, G=0$$

$$B^2 - 4AC = 16 - 4 \cdot 1 \cdot 3 = 4 > 0 \Rightarrow \text{E.D.P. hiperbólica. } \frac{dy}{dx} = \frac{4 \pm \sqrt{4}}{2} \Rightarrow \frac{dy}{dx} = 3 \text{ ou } \frac{dy}{dx} = 1 \Rightarrow y = 3x + k_1$$

$$\text{e } y = x + k_2 \Rightarrow k_1 = y - 3x \text{ e } k_2 = y - x.$$

Definimos $\xi = y - x$, $\eta = y - 3x$ que são as curvas características.

$$\xi_x = -1, \xi_y = 1, \xi_{xx} = \xi_{yy} = \xi_{xy} = 0, \xi_x^2 = 1, \xi_y^2 = 1, \eta_x = -3, \eta_y = 1, \eta_{xx} = \eta_{yy} = \eta_{xy} = 0, \eta_x^2 = 9, \eta_y^2 = 1.$$

$$A' = 1 \cdot 1 + 4 \cdot (-1) \cdot 1 + 3 \cdot 1 = 0, B' = 2 \cdot 1 \cdot (-1) \cdot (-3) + 4 \cdot (-1) \cdot 1 + 1 \cdot (-3) = 6 - 4 - 3 = -1, C' = 1 \cdot 9 + 4 \cdot (-3) \cdot 1 + 3 \cdot 1 = 0, D' = 3 \cdot (-1) + (-1) \cdot 1 = -4, E' = 3 \cdot (-3) + (-1) \cdot 1 = -10, F' = 2, G' = 0$$

$$-4u_{\xi\eta} - 4u_{\xi} - 10u_{\eta} + 2u = 0 \Rightarrow u_{\xi\eta} = \frac{2u - 4u_{\xi} - 10u_{\eta}}{4} = \frac{u - 2u_{\xi} - 5u_{\eta}}{2}, \text{ essa é a primeira forma}$$

canônica, para a segunda temos, $\alpha = \xi + \eta$ e $\beta = \xi - \eta$

$$\alpha = y - x + y - 3x = 2y - 4x, \beta = y - x - (y - 3x) = 2x$$

$$\alpha_x = -4, \alpha_y = 2, \alpha_{xx} = \alpha_{yy} = \alpha_{xy} = 0, \alpha_x^2 = 16, \alpha_y^2 = 4, \beta_x = 2, \beta_y = \beta_{xx} = \beta_{yy} = \beta_{xy} = \beta_y^2 = 0, \beta_x^2 = 4$$

$$A'' = 1 \cdot 16 + 4 \cdot (-4) \cdot 2 + 3 \cdot 4 = 28 - 32 = -4, B'' = 2 \cdot 1 \cdot (-4) \cdot 2 + 4 \cdot (-4) \cdot 0 + 2 \cdot 2 = -16 + 16 = 0$$

$$C'' = 1 \cdot 4 + 4 \cdot 2 \cdot 0 + 3 \cdot 0 = 4, D'' = 3 \cdot (-4) + (-1) \cdot 2 = -14, E'' = 3 \cdot 2 + (-1) \cdot 0 = 6$$

$$-4u_{\alpha\alpha} + 4u_{\beta\beta} - 14u_{\alpha} + 6u_{\beta} + 2u = 0 \Leftrightarrow u_{\alpha\alpha} - u_{\beta\beta} + \frac{7}{2}u_{\alpha} - \frac{3}{2}u_{\beta} - \frac{1}{2}u = 0$$

$$b) u_{xx} + 2u_{xy} + u_{yy} + 5u_x + 3u_y + u = 0 \Rightarrow A=1, B=2, C=1, D=5, E=3, F=1, G=0.$$

$$B^2 - 4AC = 4 - 4 \cdot 1 \cdot 1 = 0 \Rightarrow \text{E.D.P. parabólica. } \frac{dy}{dx} = \frac{2 \pm \sqrt{0}}{2} = 1 \Rightarrow dy = dx \Rightarrow y - x = k_1 = \xi, \text{ tomando}$$

$$\eta = y, \text{ temos } J(\xi, \eta) = \begin{vmatrix} -1 & 1 \\ 0 & 1 \end{vmatrix} = -1 \neq 0$$

$$u_x = u_{\xi} \xi_x + u_{\eta} \eta_x = -u_{\xi}; u_y = u_{\xi} \xi_y + u_{\eta} \eta_y = u_{\xi} + u_{\eta}$$

$$u_{xx} = -\frac{\partial}{\partial x} u_{\xi} = -u_{\xi\xi} \xi_x - u_{\xi\eta} \eta_x = u_{\xi\xi}; u_{yy} = \frac{\partial}{\partial y} u_{\xi} + \frac{\partial}{\partial y} u_{\eta} = u_{\xi\xi} \xi_y + u_{\xi\eta} \eta_y + u_{\eta\xi} \xi_y + u_{\eta\eta} \eta_y$$

$$= u_{\xi\xi} + u_{\xi\eta} + u_{\eta\xi} + u_{\eta\eta} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{xy} = \frac{\partial}{\partial x} u_y + \frac{\partial}{\partial x} u_y = u_{yx} \xi_x + u_{yx} \eta_x + u_{yx} \xi_x + u_{yx} \eta_x = -u_{yx} - u_{yx}$$

$$0 = u_{xx} + 2u_{xy} + u_{yy} + 5u_x + 3u_y + u = \cancel{u_{xx}} - 2\cancel{u_{xx}} - 2\cancel{u_{xy}} + \cancel{u_{xy}} + 2\cancel{u_{xy}} + u_{yy} - 5u_x + 3u_x + 3u_y + u$$

$$= u_{yy} - 2u_x + 3u_y + u = 0.$$

c) $u_{xx} - 6u_{xy} + 12u_{yy} + 4u_x - u = 0 \Rightarrow A=1, B=-6, C=12, D=4, E=0, F=-1, G=0.$

$$B^2 - 4AC = 36 - 4 \cdot 1 \cdot 12 = 36 - 48 = -12 < 0 \Rightarrow \text{E.D.P. elíptica. } \frac{dy}{dx} = \frac{-6 \pm \sqrt{-12}}{2} = \frac{-6 \pm 2i\sqrt{3}}{2} = -3 \pm i\sqrt{3}$$

$$\Rightarrow y = -(3+i\sqrt{3})x + k_1 \text{ e } y = (-3+i\sqrt{3})x + k_2 \Rightarrow \xi = y + (3+i\sqrt{3})x \text{ e } \eta = y + (3-i\sqrt{3})x$$

$$\alpha = \frac{1}{2}(\xi + \eta) = \frac{1}{2}(y + 3x + xi\sqrt{3} + y + 3x - xi\sqrt{3}) = \frac{1}{2}(2y + 6x) = y + 3x.$$

$$\beta = \frac{1}{2i}(\xi - \eta) = \frac{1}{2i}(y + 3x + xi\sqrt{3} - y - 3x + xi\sqrt{3}) = x\sqrt{3}$$

$$u_x = u_\alpha \alpha_x + u_\beta \beta_x = 3u_\alpha + \sqrt{3}u_\beta; \quad u_y = u_\alpha \alpha_y + u_\beta \beta_y = u_\alpha$$

$$u_{xx} = 3 \frac{\partial}{\partial x} u_\alpha + \sqrt{3} \frac{\partial}{\partial x} u_\beta = 3(u_{\alpha\alpha} \alpha_x + u_{\alpha\beta} \beta_x) + \sqrt{3}(u_{\beta\alpha} \alpha_x + u_{\beta\beta} \beta_x) = 3(u_{\alpha\alpha} 3 + \sqrt{3}u_{\alpha\beta}) + \sqrt{3}(u_{\beta\alpha} 3 + \sqrt{3}u_{\beta\beta})$$

$$= 9u_{\alpha\alpha} + 3\sqrt{3}u_{\alpha\beta} + 3\sqrt{3}u_{\beta\alpha} + 3u_{\beta\beta} = 9u_{\alpha\alpha} + 3u_{\beta\beta} + 6\sqrt{3}u_{\alpha\beta}$$

$$u_{yy} = \frac{\partial}{\partial y} u_\alpha = u_{\alpha\alpha} \alpha_y + u_{\alpha\beta} \beta_y = u_{\alpha\alpha}$$

$$u_{xy} = \frac{\partial}{\partial x} u_\alpha = u_{\alpha\alpha} \alpha_x + u_{\alpha\beta} \beta_x = 3u_{\alpha\alpha} + \sqrt{3}u_{\alpha\beta}$$

$$0 = u_{xx} - 6u_{xy} + 12u_{yy} + 4u_x - u = 9u_{\alpha\alpha} + 3u_{\beta\beta} + 6\sqrt{3}u_{\alpha\beta} - \cancel{18u_{\alpha\alpha}} - \cancel{6\sqrt{3}u_{\alpha\beta}} + 12u_{\alpha\alpha} + 12u_\alpha + 4\sqrt{3}u_\beta - u$$

= ; (2) Encontre a forma canônica das seguintes EDPs:

(a) $u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 0,$

(b) $u_{xx} + 2u_{xy} + u_{yy} + 5u_x + 3u_y + u = 0,$

(c) $u_{xx} - 6u_{xy} + 12u_{yy} + 4u_x - u = 0.$

$$+ \frac{4}{\sqrt{3}}u_\beta - \frac{1}{3}u = 0$$

↳ no gabarito está $-\frac{4}{\sqrt{3}}$

→ refazendo utilizando outra abordagem.

a) $u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 0$, $B^2 - 4AC = 16 - 4 \cdot 1 \cdot 3 = 4 > 0 \Rightarrow$ E.D.P. hiperbólica,

$$\frac{dy}{dx} = \frac{4 \pm \sqrt{4}}{2} = 2 \pm 1 \Rightarrow \begin{cases} dy = 3dx \\ dy = dx \end{cases} \Rightarrow \begin{cases} y = 3x + k_2 \\ y = x + k_1 \end{cases} \Rightarrow \begin{cases} \eta = y - 3x \\ \xi = y - x \end{cases}$$

$$\alpha = \xi + \eta = y - x + y - 3x = 2y - 4x, \quad \beta = \xi - \eta = y - x - y + 3x = 2x$$

$$u_x = u_\alpha \alpha_x + u_\beta \beta_x = -4u_\alpha + 2u_\beta$$

$$u_y = u_\alpha \alpha_y + u_\beta \beta_y = 2u_\alpha$$

$$u_{xx} = -4 \frac{\partial}{\partial x} u_\alpha + 2 \frac{\partial}{\partial x} u_\beta = -4(u_{\alpha\alpha} \alpha_x + u_{\alpha\beta} \beta_x) + 2(u_{\beta\alpha} \alpha_x + u_{\beta\beta} \beta_x) = -4(-4u_{\alpha\alpha} + 2u_{\alpha\beta}) + 2(-4u_{\beta\alpha} + 2u_{\beta\beta})$$

$$= 16u_{\alpha\alpha} - 8u_{\alpha\beta} - 8u_{\beta\alpha} + 4u_{\beta\beta} = 16u_{\alpha\alpha} + 4u_{\beta\beta} - 16u_{\alpha\beta}$$

$$u_{yy} = 2 \frac{\partial}{\partial y} u_\alpha = 2(u_{\alpha\alpha} \alpha_y + u_{\alpha\beta} \beta_y) = 2(u_{\alpha\alpha} \cdot 2) = 4u_{\alpha\alpha}$$

$$u_{xy} = 2 \frac{\partial}{\partial x} u_\alpha = 2(u_{\alpha\alpha} \alpha_x + u_{\alpha\beta} \beta_x) = 2(-4u_{\alpha\alpha} + 2u_{\alpha\beta}) = -8u_{\alpha\alpha} + 4u_{\alpha\beta}$$

$$0 = u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 16u_{\alpha\alpha} + 4u_{\beta\beta} - 16u_{\alpha\beta} - 32u_{\alpha\alpha} + 16u_{\alpha\beta} + 12u_{\alpha\alpha} - 12u_\alpha + 6u_\beta - 2u_\alpha$$

$$+ 2u = -4u_{\alpha\alpha} + 4u_{\beta\beta} - 14u_\alpha + 6u_\beta + 2u = u_{\alpha\alpha} - u_{\beta\beta} + \frac{7}{2}u_\alpha - \frac{3}{2}u_\beta - \frac{1}{2}u = 0$$

3)

a) $0 = u_{xx} + y u_{yy} + \frac{1}{2} u_y \Rightarrow A=1, C=y, E=\frac{1}{2}, B=D=F=G=0. (y < 0)$

$$B^2 - 4AC = -4y > 0 \text{ pois } y < 0, \text{ logo a E.D.P. é hiperbólica.}$$

$$\frac{dy}{dx} = \frac{\pm \sqrt{-4y}}{2} = \pm \sqrt{-y} \Rightarrow \begin{cases} dy = -\sqrt{-y} dx \\ dy = \sqrt{-y} dx \end{cases}, \quad \int \frac{1}{(-y)^{1/2}} dy = \int (-y)^{-1/2} dy \stackrel{y \rightarrow -y}{=} \int y^{-1/2} dy = -\frac{1}{1/2} y^{1/2} + C_1$$

$$= -2\sqrt{-y} + C_1 = -\int dx = -x + C_2 \Rightarrow \eta = x - 2\sqrt{-y}$$

$$-2\sqrt{-y} + C_3 = x + C_4 \Rightarrow \xi = x + 2\sqrt{-y}$$

$$\alpha = \xi + \eta = x + 2\sqrt{-y} + x - 2\sqrt{-y} = 2x, \quad \beta = \xi - \eta = x + 2\sqrt{-y} - x + 2\sqrt{-y} = 4\sqrt{-y}$$

$$u_x = u_\alpha \alpha_x + u_\beta \beta_x = 2u_\alpha, \quad u_y = u_\alpha \alpha_y + u_\beta \beta_y = -2(-y)^{1/2} u_\beta$$

$$u_{xx} = 2 \frac{\partial}{\partial x} u_x = 2(u_{\alpha\alpha}\alpha_x + u_{\alpha\beta}\beta_x) = 4u_{\alpha\alpha}$$

$$u_{yy} = -2 \frac{\partial}{\partial y} (-y)^{\frac{1}{2}} u_\beta = -2 \left(\frac{1}{2} (-y)^{-\frac{1}{2}} u_\beta + (-y)^{\frac{1}{2}} \frac{\partial}{\partial y} u_\beta \right) = -(-y)^{-\frac{1}{2}} u_\beta - 2(-y)^{\frac{1}{2}} (u_{\beta\alpha}\alpha_y + u_{\beta\beta}\beta_y) = -(-y)^{-\frac{1}{2}} u_\beta - 2(-y)^{\frac{1}{2}} (2(-y)^{\frac{1}{2}} u_{\beta\beta})$$

$$= -(-y)^{-\frac{1}{2}} u_\beta - 4y u_{\beta\beta}$$

$$0 = u_{xx} + y u_{yy} + \frac{1}{2} u_y = 4u_{\alpha\alpha} - y(-y)^{-\frac{1}{2}} u_\beta - 4u_{\beta\beta} - (-y)^{\frac{1}{2}} u_\beta = 4u_{\alpha\alpha} + (-y)^{-\frac{1}{2}} u_\beta - 4u_{\beta\beta} - (-y)^{\frac{1}{2}} u_\beta$$

$$= u_{\alpha\alpha} - u_{\beta\beta} = 0 \rightarrow \text{muito tempo achar essa solução}$$

$$\xi = x + 2\sqrt{y}, \eta = x - 2\sqrt{y}, u_x = u_\xi \xi_x + u_\eta \eta_x = u_\xi + u_\eta; u_y = u_\xi \xi_y + u_\eta \eta_y = -(-y)^{-\frac{1}{2}} u_\xi + (-y)^{-\frac{1}{2}} u_\eta$$

$$\xi_y = -(-y)^{-\frac{1}{2}}, \eta_y = (-y)^{-\frac{1}{2}}, \xi_y^2 = (-y)^{-1} = -\bar{y}^{-1}, \eta_y^2 = (-y)^{-1} = -\bar{y}^{-1}$$

$$u_{xx} = \frac{\partial}{\partial x} u_\xi + \frac{\partial}{\partial x} u_\eta = u_{\xi\xi} \xi_x + u_{\xi\eta} \eta_x + u_{\eta\xi} \xi_x + u_{\eta\eta} \eta_x = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{yy} = \frac{\partial}{\partial y} (-y)^{-\frac{1}{2}} (u_\eta - u_\xi) = \frac{1}{2} (-y)^{-\frac{3}{2}} (u_\eta - u_\xi) + (-y)^{-\frac{1}{2}} \frac{\partial}{\partial y} (u_\eta - u_\xi) = \frac{1}{2} (-y)^{-\frac{3}{2}} (u_\eta - u_\xi) + (-y)^{-\frac{1}{2}} (u_{\eta\xi} \xi_y + u_{\eta\eta} \eta_y - u_{\xi\xi} \xi_y - u_{\xi\eta} \eta_y)$$

$$= \frac{1}{2} (-y)^{-\frac{3}{2}} (u_\eta - u_\xi) + (-y)^{-\frac{1}{2}} (-u_{\eta\xi} (-y)^{-\frac{1}{2}} + u_{\eta\eta} (-y)^{-\frac{1}{2}} + u_{\xi\xi} (-y)^{-\frac{1}{2}} - u_{\xi\eta} (-y)^{-\frac{1}{2}})$$

$$= \frac{1}{2} (-y)^{-\frac{3}{2}} (u_\eta - u_\xi) + u_{\eta\xi} \bar{y}^{-1} - u_{\eta\eta} \bar{y}^{-1} - u_{\xi\xi} \bar{y}^{-1} + u_{\xi\eta} \bar{y}^{-1} = \frac{1}{2} (-y)^{-\frac{3}{2}} (u_\eta - u_\xi) + \frac{1}{y} (2u_{\xi\eta} - u_{\eta\eta} - u_{\xi\xi})$$

$$0 = u_{xx} + y u_{yy} + \frac{1}{2} u_y = \cancel{u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}} - \cancel{\frac{1}{2} (-y)^{-\frac{3}{2}} (u_\eta - u_\xi)} + \cancel{2u_{\xi\eta} - u_{\eta\eta} - u_{\xi\xi}} + \frac{1}{2} (-y)^{-\frac{3}{2}} (u_\eta - u_\xi)$$

$$= 4u_{\xi\eta} = 0 \Rightarrow u_{\xi\eta} = 0 \Rightarrow u_\xi = \phi(\xi) \Rightarrow u(\xi, \eta) = \phi(\xi) + \psi(\eta)$$

↑ pag 176.

$$u(\xi(x, y), \eta(x, y)) = u(x, y) = \phi(x + 2\sqrt{y}) + \psi(x - 2\sqrt{y})$$

b) $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 0 \Rightarrow A=x^2, B=2xy, C=y^2$.

$B^2 - 4AC = 4x^2y^2 - 4x^2y^2 = 0 \Rightarrow$ E.D.P parabólica. $\frac{dy}{dx} = \frac{2xy}{2x^2} = \frac{y}{x} \Rightarrow \frac{1}{y} dy = \frac{1}{x} dx \Rightarrow \ln|y| = \ln|x| + C$

$\Rightarrow y = xk \Rightarrow k = \frac{y}{x} = \xi$, tomando $\eta = y$, $J(\xi, \eta) = \begin{vmatrix} -\frac{1}{x^2}\eta & \frac{1}{x} \\ 0 & y \end{vmatrix} = -\left(\frac{1}{x}\eta\right)^2 \neq 0 \Leftrightarrow y \neq 0$,

$u_x = u_\xi \xi_x + u_\eta \eta_x = -\frac{1}{x^2} \eta u_\xi$, $u_y = u_\xi \xi_y + u_\eta \eta_y = \frac{1}{x} u_\xi + u_\eta$

$u_{xx} = -\eta \frac{\partial}{\partial x} \frac{1}{x^2} u_\xi = -\eta \left(-\frac{2}{x^3} u_\xi + \frac{1}{x^2} \frac{\partial}{\partial x} u_\xi \right) = \frac{2\eta}{x^3} u_\xi - \frac{1}{x^2} \eta (u_{\xi\xi} \xi_x + u_{\xi\eta} \eta_x) = \frac{1}{x^3} 2\eta u_\xi - \frac{1}{x^2} \eta \left(-\frac{1}{x^2} \eta u_{\xi\xi} \right)$

$= \frac{1}{x^3} 2\eta u_\xi + \frac{1}{x^4} \eta^2 u_{\xi\xi}$

$u_{yy} = \frac{1}{x} \frac{\partial}{\partial y} u_\xi + \frac{\partial}{\partial y} u_\eta = \frac{1}{x} (u_{\xi\xi} \xi_y + u_{\xi\eta} \eta_y) + u_{\eta\xi} \xi_y + u_{\eta\eta} \eta_y = \frac{1}{x} \left(u_{\xi\xi} \frac{1}{x} + u_{\xi\eta} \eta \right) + u_{\eta\xi} \frac{1}{x} + u_{\eta\eta}$

$= \frac{1}{x^2} u_{\xi\xi} + \frac{2}{x} u_{\xi\eta} + u_{\eta\eta}$

$u_{xy} = \frac{\partial}{\partial x} \left(\frac{1}{x} u_\xi \right) + \frac{\partial}{\partial x} u_\eta = -\frac{1}{x^2} u_\xi + \frac{1}{x} (u_{\xi\xi} \xi_x + u_{\xi\eta} \eta_x) + u_{\eta\xi} \xi_x + u_{\eta\eta} \eta_x = -\frac{1}{x^2} u_\xi + \frac{1}{x} \left(-\frac{1}{x^2} \eta u_{\xi\xi} \right) - \frac{1}{x^2} \eta u_{\xi\eta}$

$= -\frac{1}{x^2} u_\xi - \frac{1}{x^3} \eta u_{\xi\xi} - \frac{1}{x^2} \eta u_{\xi\eta}$

~~$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = \frac{2\eta}{x^3} u_\xi + \frac{1}{x^4} \eta^2 u_{\xi\xi} - \frac{2\eta}{x^3} u_\xi - \frac{2}{x^2} \eta^2 u_{\xi\xi} - \frac{2}{x^2} \eta^2 u_{\xi\eta} + \frac{1}{x^2} \eta^2 u_{\xi\xi} + \frac{2}{x^2} \eta^2 u_{\xi\eta} + y^2 u_{\eta\eta}$~~

$= y^2 u_{\eta\eta} = \eta^2 u_{\eta\eta}$, como $y \neq 0$, $\eta \neq 0$, luego $u_{\eta\eta} = 0 \Rightarrow u(\xi, \eta) = \phi(\xi) + \eta \psi(\xi)$

$\Rightarrow u(x, y) = \phi\left(\frac{y}{x}\right) + y \psi\left(\frac{y}{x}\right)$

c) $u_{xx} + \frac{10}{3} u_{xy} + u_{yy} = 0 \Rightarrow A=1, B=\frac{10}{3}, C=1$. $B^2 - 4AC = \frac{100}{9} - 4 = \frac{100-36}{9} = \frac{64}{9} > 0 \Rightarrow$ E.D.P hiperbólica.

$\frac{dy}{dx} = \frac{10/3 \pm \sqrt{64/9}}{2} = \frac{10/3 \pm 8/3}{2} \Rightarrow \frac{dy}{dx} = \frac{18}{6} = 3$ e $\frac{dy}{dx} = \frac{2}{6} = \frac{1}{3} \Rightarrow y = 3x + k_1$ e $y = \frac{1}{3}x + k_2$

$$\Rightarrow \xi = y - 3x \text{ e } \eta = y - \frac{1}{3}x$$

$$u_x = u_\xi \xi_x + u_\eta \eta_x = -3u_\xi - \frac{1}{3}u_\eta, \quad u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$u_{xx} = -3 \frac{\partial}{\partial x} u_\xi - \frac{1}{3} \frac{\partial}{\partial x} u_\eta = -3(u_{\xi\xi} \xi_x + u_{\xi\eta} \eta_x) - \frac{1}{3}(u_{\eta\xi} \xi_x + u_{\eta\eta} \eta_x) = -3(-3u_{\xi\xi} - \frac{1}{3}u_{\xi\eta}) - \frac{1}{3}(-3u_{\eta\xi} - \frac{1}{3}u_{\eta\eta}) = 9u_{\xi\xi} + u_{\xi\eta} + u_{\eta\xi} + \frac{1}{9}u_{\eta\eta} = 9u_{\xi\xi} + 2u_{\xi\eta} + \frac{1}{9}u_{\eta\eta}$$

$$u_{yy} = \frac{\partial}{\partial y} u_\xi + \frac{\partial}{\partial y} u_\eta = u_{\xi\xi} \xi_y + u_{\xi\eta} \eta_y + u_{\eta\xi} \xi_y + u_{\eta\eta} \eta_y = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{xy} = \frac{\partial}{\partial x} u_\xi + \frac{\partial}{\partial x} u_\eta = u_{\xi\xi} \xi_x + u_{\xi\eta} \eta_x + u_{\eta\xi} \xi_x + u_{\eta\eta} \eta_x = -3u_{\xi\xi} - \frac{1}{3}u_{\xi\eta} - 3u_{\eta\xi} - \frac{1}{3}u_{\eta\eta} = -3u_{\xi\xi} - \frac{10}{3}u_{\xi\eta} - \frac{1}{3}u_{\eta\eta}$$

$$0 = u_{xx} + \frac{10}{3}u_{xy} + u_{yy} = \cancel{9u_{\xi\xi}} + \cancel{2u_{\xi\eta}} + \cancel{\frac{1}{9}u_{\eta\eta}} - \cancel{10u_{\xi\xi}} - \cancel{\frac{100}{3}u_{\xi\eta}} - \cancel{\frac{10}{9}u_{\eta\eta}} + \cancel{u_{\xi\xi}} + \cancel{2u_{\xi\eta}} + \cancel{u_{\eta\eta}} = \frac{18u_{\xi\eta} - 100u_{\xi\eta} + 18u_{\xi\eta}}{9} = -\frac{64}{9}u_{\xi\eta} = 0 \Rightarrow u_{\xi\eta} = 0 \Rightarrow u(\xi, \eta) = \phi(\xi) + \psi(\eta)$$

$$\Rightarrow u(x, y) = \phi(y - 3x) + \psi(y - \frac{1}{3}x)$$

$$d) 3u_{xx} + 5u_{xy} + 2u_{yy} + u_x + u_y = -\frac{1}{3} \Rightarrow A=3, B=5, C=2, D=E=1, F=0, G=-\frac{1}{3}$$

$$B^2 - 4AC = 25 - 4 \cdot 3 \cdot 2 = 1 > 0 \Rightarrow \text{E.D.P. hiperbolica. } \frac{dy}{dx} = \frac{5 \pm 1}{6} \Rightarrow \frac{dy}{dx} = 1 \text{ e } \frac{dy}{dx} = \frac{2}{3} \Rightarrow y = x + k_1$$

$$\text{e } y = \frac{2}{3}x + k_2 \Rightarrow \xi = y - x, \eta = y - \frac{2}{3}x$$

$$u_x = u_\xi \xi_x + u_\eta \eta_x = -u_\xi - \frac{2}{3}u_\eta, \quad u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$u_{xx} = -\frac{\partial}{\partial x} u_\xi - \frac{2}{3} \frac{\partial}{\partial x} u_\eta = -(u_{\xi\xi} \xi_x + u_{\xi\eta} \eta_x) - \frac{2}{3}(u_{\eta\xi} \xi_x + u_{\eta\eta} \eta_x) = -(-u_{\xi\xi} - \frac{2}{3}u_{\xi\eta}) - \frac{2}{3}(-u_{\eta\xi} - \frac{2}{3}u_{\eta\eta})$$

$$= u_{\xi\xi} + \frac{4}{3}u_{\xi\eta} + \frac{4}{9}u_{\eta\eta}$$

$$u_{\eta\eta} = \frac{\partial}{\partial \eta} u_{\xi} + \frac{\partial}{\partial \eta} u_{\eta} = u_{\xi\xi}\xi_{\eta} + u_{\xi\eta}\eta_{\eta} + u_{\eta\xi}\xi_{\eta} + u_{\eta\eta}\eta_{\eta} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{\xi\eta} = \frac{\partial}{\partial \xi} u_{\xi} + \frac{\partial}{\partial \xi} u_{\eta} = u_{\xi\xi}\xi_{\xi} + u_{\xi\eta}\eta_{\xi} + u_{\eta\xi}\xi_{\xi} + u_{\eta\eta}\eta_{\xi} = -u_{\xi\xi} - \frac{2}{3}u_{\xi\eta} - u_{\eta\xi} - \frac{2}{3}u_{\eta\eta} = -u_{\xi\xi} - \frac{5}{3}u_{\xi\eta} - \frac{2}{3}u_{\eta\eta}$$

$$-\frac{1}{3} = 3u_{\xi\xi} + 5u_{\xi\eta} + 2u_{\eta\eta} + u_{\xi} + u_{\eta} = 3u_{\xi\xi} + 4u_{\xi\eta} + \frac{4}{3}u_{\eta\eta} - 5u_{\xi\xi} - \frac{25}{3}u_{\xi\eta} - \frac{10}{3}u_{\eta\eta} + 2u_{\xi\xi} + 4u_{\xi\eta} + 2u_{\eta\eta}$$

$$-\frac{1}{3}u_{\xi} - \frac{2}{3}u_{\eta} + u_{\xi} + u_{\eta} = \frac{(24-25)u_{\xi\eta}}{3} + \frac{(4-10+6)u_{\eta\eta}}{3} + \frac{(3-2)u_{\eta}}{3} = -\frac{1}{3}u_{\xi\eta} + \frac{1}{3}u_{\eta} = -\frac{1}{3} \Rightarrow u_{\xi\eta} - u_{\eta} = 1$$

$$\Rightarrow u_{\xi\eta} = 1 + u_{\eta} \cdot \frac{\partial^2}{\partial \xi \partial \eta} u = 1 + \frac{\partial}{\partial \eta} u, \quad e^{-\int d\xi} = e^{-\xi+c} = e^{-\xi}C, \quad C e^{-\xi} u_{\xi\eta} - C e^{-\xi} u_{\eta} = C e^{-\xi} \Rightarrow e^{-\xi} u_{\xi\eta} - e^{-\xi} u_{\eta} = e^{-\xi}$$

$$\Rightarrow \frac{\partial}{\partial \xi} (e^{-\xi} u_{\eta}) = e^{-\xi} \Rightarrow e^{-\xi} u_{\eta} = -e^{-\xi} + \phi(\eta) \Rightarrow u_{\eta} = -1 + C(\eta) e^{\xi} \Rightarrow u(\xi, \eta) = -\eta + e^{\xi} \int C(\eta) d\eta + \psi(\xi)$$

$$= \psi(\xi) + \phi(\eta) e^{\xi} - \eta \Rightarrow \psi(y-x) + \phi(y-\frac{2}{3}x) e^{y-x} - (y-\frac{2}{3}x) = u(x, y).$$

4)

$$a) \begin{cases} u_{tt} - c^2 u_{xx} = 0 & B^2 - 4AC = -4(1)(-c^2) = 4c^2, \quad \frac{dt}{dx} = \frac{\pm \sqrt{4c^2}}{2(-c^2)} = \frac{\pm 2c}{2(-c^2)} = \mp \frac{1}{c} \Rightarrow \begin{cases} \xi = x + tc \\ \eta = x - tc \end{cases} \\ u(x, 0) = 0 \\ u_t(x, 0) = x \end{cases} \quad \underbrace{B^2 - 4AC > 0, \forall c \neq 0.}_{\text{E.D.P. hiperbólica, pela solução de D'Alembert,}}$$

$$u(\xi, \eta) = \phi(\xi) + \psi(\eta) \Rightarrow u(x, y) = \phi(x+tc) + \psi(x-tc)$$

$$u(x, 0) = \phi(x) + \psi(x); \quad u_t(x, y) = c\phi'(x+tc) - c\psi'(x-tc) \Rightarrow u_t(x, 0) = c\phi'(x) - c\psi'(x) = x$$

$$\int u_t(x, 0) dx = c \int \phi'(x) dx - c \int \psi'(x) dx = c\phi(x) - c\psi(x) + C' = \int x dx = \frac{1}{2}x^2 \Big|_{x_0}^x = \frac{1}{2}(x^2 - x_0^2) \\ \Rightarrow c(\phi(x) - \psi(x)) = \frac{1}{2}(x^2 - x_0^2) + K \Rightarrow \phi(x) - \psi(x) = \frac{1}{2c}(x^2 - x_0^2) + K, \quad K \text{ e } x_0 \text{ são constantes.}$$

$$\begin{cases} \phi(x) + \psi(x) = 0 \\ \phi(x) - \psi(x) = \frac{1}{2c}(x^2 - x_0^2) + K \end{cases} \Rightarrow \phi(x) = \frac{1}{4c}(x^2 - x_0^2) + \frac{K}{4} \text{ e } \psi(x) = -\frac{1}{4c}(x^2 - x_0^2) - \frac{K}{4}$$

$$u(x,t) = \phi(x+tc) + \psi(x-tc) = \frac{1}{4c} [(x+tc)^2 - x_0^2 - (x-tc)^2 + x_0^2] = \frac{1}{4c} [\cancel{x^2} + 2xtc + \cancel{t^2c^2} - \cancel{x^2} + 2xtc - \cancel{t^2c^2}]$$

$$= \frac{4xtc}{4c} = xt, \therefore u(x,t) = xt.$$

b)
$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{novamente, } \xi = x+tc \text{ e } \eta = x-tc, \text{ ainda} \\ u(x,0) = \sin x & u(x,t) = \phi(x+tc) + \psi(x-tc) \text{ e } u_t(x,t) = c\phi'(x+tc) - c\psi'(x-tc) \\ u_t(x,0) = 1 & u(x,0) = \phi(x) + \psi(x) = \sin x, \quad u_t(x,0) = c\phi'(x) - c\psi'(x) = 1 \end{cases}$$

$$\int u_t(x,0) dx = c\phi(x) - c\psi(x) = \int_{x_0}^x dx + K = \phi(x) - \psi(x) = \frac{1}{c}(x-x_0) + K, \quad x_0 \in \mathbb{R} \text{ cte.}$$

$$\begin{cases} \phi(x) + \psi(x) = \sin x \\ \phi(x) - \psi(x) = \frac{1}{c}(x-x_0) + K \end{cases} \Rightarrow \phi(x) = \frac{1}{2} \left(\sin x + \frac{1}{c}(x-x_0) + K \right) \text{ e } \psi(x) = \frac{1}{2} \left(\sin x - \frac{1}{c}(x-x_0) - K \right)$$

$$u(x,t) = \frac{1}{2} [\sin(x+tc) + \sin(x-tc)] + \frac{1}{2c} [x+tc-x_0 - (x-tc) + x_0] + \frac{K}{2} - \frac{K}{2}$$

$$= \frac{1}{2} [\cancel{\sin x \cos tc} + \cancel{\sin tc \cos x} + \sin x \cos tc - \sin tc \cos x] + \frac{2tc}{2c} = \sin x \cos tc + t$$

$$\therefore u(x,t) = t + \sin x \cos ct.$$

c)
$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \xi = x+tc, \eta = x-tc; \quad u(x,t) = \phi(x+tc) + \psi(x-tc) \\ u(x,0) = x & u(x,0) = \phi(x) + \psi(x) = x, \quad u_t(x,0) = c\phi'(x) - c\psi'(x) \Rightarrow c\phi(x) - c\psi(x) = \frac{1}{3}x^3 \Big|_{x_0}^x + K \\ u_t(x,0) = x^2 & \Rightarrow \phi(x) - \psi(x) = \frac{1}{3c}(x^3 - x_0^3) + K \end{cases}$$

$$\begin{cases} \phi(x) + \psi(x) = x \\ \phi(x) - \psi(x) = \frac{1}{3c}(x^3 - x_0^3) + K \end{cases} \Rightarrow \phi(x) = \frac{1}{2} \left(x + \frac{1}{3c}(x^3 - x_0^3) + K \right) \text{ e } \psi(x) = \frac{1}{2} \left(x - \frac{1}{3c}(x^3 - x_0^3) - K \right)$$

$$u(x,t) = \frac{1}{2} [x+tc + x-tc] + \frac{1}{6c} [(x+tc)^3 - x_0^3 - (x-tc)^3 + x_0^3] + \frac{K}{2} - \frac{K}{2}$$

$$= x + \frac{1}{6c} [\cancel{x^3} + 3x^2ct + 3xc^2t^2 + \cancel{c^3t^3} - \cancel{x^3} + 3x^2ct - 3xc^2t^2 + \cancel{c^3t^3}] = x + \frac{6x^2ct + 2c^3t^3}{6c} = x + x^2t + \frac{1}{3}c^2t^3$$

$$u(x,t) = x + x^2 t + \frac{1}{3} c^2 t^3$$

d)
$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \xi = x + ct, \quad \eta = x - ct, \quad u(x,t) = \phi(x+ct) + \psi(x-ct) \\ u(x,0) = \cos x & u(x,0) = \phi(x) + \psi(x) = \cos x, \quad u_t(x,t) = c\phi'(x+ct) - c\psi'(x-ct) \\ u_t(x,0) = c \sin x & u_t(x,0) = c\phi'(x) - c\psi'(x) = c \sin x \end{cases}$$

$$c\phi(x) - c\psi(x) + C'' = C \int_{x_0}^x \sin x = -C \cos x \Big|_{x_0}^x = C(\cos x_0 - \cos x) \Rightarrow \phi(x) - \psi(x) = \cos x_0 - \cos x + K$$

K_1 e K_2 constantes.

$$\begin{cases} \phi(x) + \psi(x) = \cos x \\ \phi(x) - \psi(x) = \cos x_0 - \cos x + K \end{cases} \Rightarrow \phi(x) = \frac{1}{2} [\cos x + \cos x_0 - \cos x + K]$$

$$\text{e } \psi(x) = \frac{1}{2} [\cos x - (\cos x_0 - \cos x) - K]$$

$$u(x,t) = \frac{1}{2} [\cos(x+ct) + \cos(x-ct)] + \frac{1}{2} [\cos x_0 - \cos(x+ct) - \cos x_0 + \cos(x-ct)] + \frac{K}{2} - \frac{K}{2}$$

$$= \frac{1}{2} [\cancel{\cos x \cos ct} - \cancel{\sin x \sin ct} + \cancel{\cos x \cos ct} + \cancel{\sin x \sin ct}] + \frac{1}{2} [\cancel{-\cos x \cos ct} + \cancel{\sin x \cos ct} + \cancel{\cos x \cos ct} + \cancel{\sin x \sin ct}] = \cos x \cos ct + \sin x \sin ct = \cos(x-ct)$$

$$\therefore u(x,t) = \cos(x-ct)$$

e)
$$\begin{cases} u_{tt} - c^2 u_{xx} = x \\ u(x,0) = 0 \\ u_t(x,0) = 1 \end{cases}$$

ciais $u(x,0) = \phi(x)$ e $u_t(x,0) = \psi(x)$, segue que

$$u(x,t) = \frac{1}{2} (\phi(x+ct) + \phi(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(x') dx' + \frac{1}{2} \iint_R h(x,t) dR. \quad (6.18) \quad \rightarrow \text{pág. 195}$$

$$u_{tt} - c^2 u_{xx} = x \Rightarrow \frac{1}{c^2} u_{tt} - u_{xx} = \frac{x}{c^2}$$

$$u(x,t) = \frac{1}{2c} \int_{x-ct}^{x+ct} dx + \frac{1}{2} \int_0^{ct} \int_{x+ct-z}^{x+ct-z} \frac{z}{c^2} dz = \frac{1}{2c} 2ct + \frac{1}{4c^2} \int_0^{ct} z^2 \Big|_{x-ct+z}^{x+ct-z} dz = t + \frac{1}{4c^2} \int_0^{ct} 4xct - 4xz dz$$

$$= t + \frac{1}{4c^2} (4xct \cdot ct - 2xc^2 t^2) = t + \frac{1}{4c^2} 2xc^2 t^2 = t + \frac{xt^2}{2} \quad \therefore u(x,t) = t + \frac{xt^2}{2}$$

f)
$$\begin{cases} u_{tt} - c^2 u_{xx} = e^x \Rightarrow \frac{1}{c^2} u_{tt} - u_{xx} = \frac{e^x}{c^2} \\ u(x,0) = 0 \\ u_t(x,0) = 1 \end{cases}$$

$$u(x,t) = \frac{1}{2c} \int_{x-ct}^{x+ct} dx + \frac{1}{2} \int_0^{ct} \int_{x-ct+z}^{x+ct-z} \frac{e^{\xi}}{c^2} d\xi dz = \frac{1}{2c} ct + \frac{1}{2c^2} \int_0^{ct} e^{\xi} \Big|_{x-ct+z}^{x+ct-z} dz$$

$$= t + \frac{1}{2c^2} \int_0^{ct} e^{x+ct-z} - e^{x-ct+z} dz = t + \frac{1}{2c^2} [e^{x+ct}(-e^{-z}) \Big|_0^{ct} - e^{x-ct}(e^z) \Big|_0^{ct}] = t + \frac{1}{2c^2} [e^{x+ct}(1-e^{-ct})$$

$$+ e^{x-ct}(1-e^{ct})] = t + \frac{1}{2c^2} [e^{x+ct} - e^x + e^{x-ct} - e^x] = t + \frac{1}{2c^2} [e^{x+ct} + e^{x-ct} - 2e^x] = u(x,t)$$

g)
$$\begin{cases} u_{tt} - c^2 u_{xx} = x e^{-t} \Rightarrow \frac{1}{c^2} u_{tt} - u_{xx} = \frac{x e^{-t}}{c^2} \\ u(x,0) = 1 \\ u_t(x,0) = 0 \end{cases}$$

$$u(x,t) = \frac{1}{2} [1+1] + \frac{1}{2} \int_0^{ct} \int_{x-ct+z}^{x+ct-z} \frac{\xi e^{-t}}{c^2} d\xi dz = 1 + \frac{1}{2c^2} \int_0^{ct} e^{-t} \frac{1}{2} \xi^2 \Big|_{x-ct+z}^{x+ct-z} dz$$

$$= 1 + \frac{1}{4c^2} \int_0^{ct} e^{-t} (4xct - 4xz) dz = 1 + \frac{1}{c^2} [ctx \int_0^{ct} e^{-t} dz - x \underbrace{\int_0^{ct} e^{-t} z dz}_{(*)}]$$

(*)
$$= -e^{-t} z \Big|_0^{ct} + \int_0^{ct} e^{-t} dz = -e^{-ct} - e^{-t} \Big|_0^{ct} = -ct e^{-ct} - e^{-ct} + 1$$

así mismo
$$1 + \frac{1}{c^2} [-ctx e^{-t} \Big|_0^{ct} - x(1-ct e^{-ct} - e^{-ct})] = 1 + \frac{x}{c^2} [ct(1-e^{-ct}) - 1 + ct e^{-ct} + e^{-ct}]$$

$$= 1 + \frac{x}{c^2} [ct - ct e^{-ct} + ct e^{-ct} + e^{-ct} - 1] = 1 + \frac{x}{c^2} [ct + e^{-ct} - 1] = u(x,t)$$

h)
$$\begin{cases} u_{xx} + 2u_{xt} - 3u_{tt} = 0 \Rightarrow A=1, B=2, C=-3, \text{ resto } = 0. & B^2 - 4AC = 4 - 4 \cdot 1 \cdot (-3) = 16 > 0 \\ u(x,0) = \sin x & \text{E.D.P. hiperbólica.} \\ u_t(x,0) = 1 \end{cases}$$

$$\frac{dt}{dx} = \frac{2 \pm 4}{2} \Rightarrow \begin{cases} \frac{t}{3} = x + K_2 \\ t = -x + K_1 \end{cases} \Rightarrow \begin{cases} \eta = x - t/3 \\ \xi = x + t \end{cases}$$

$$u_x = u_{\xi} \xi_x + u_{\eta} \eta_x = u_{\xi} + u_{\eta} \quad u_t = u_{\xi} \xi_t + u_{\eta} \eta_t = u_{\xi} - \frac{1}{3} u_{\eta}$$

$$u_{xx} = u_{\xi\xi} \xi_x + u_{\xi\eta} \eta_x + u_{\eta\xi} \xi_x + u_{\eta\eta} \eta_x = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{tt} = u_{\xi\xi}\xi_t + u_{\xi\eta}\eta_t - \frac{1}{3}[u_{\eta\xi}\xi_t + u_{\eta\eta}\eta_t] = u_{\xi\xi} - \frac{1}{3}u_{\xi\eta} - \frac{1}{3}[u_{\eta\xi} - \frac{1}{3}u_{\eta\eta}] = u_{\xi\xi} - \frac{2}{3}u_{\xi\eta} + \frac{1}{9}u_{\eta\eta}$$

$$u_{xt} = u_{\xi\xi}\xi_x + u_{\xi\eta}\eta_x - \frac{1}{3}[u_{\eta\xi}\xi_x + u_{\eta\eta}\eta_x] = u_{\xi\xi} + u_{\xi\eta} - \frac{1}{3}u_{\eta\xi} - \frac{1}{3}u_{\eta\eta} = u_{\xi\xi} + \frac{2}{3}u_{\xi\eta} - \frac{1}{3}u_{\eta\eta}$$

$$0 = u_{xx} + 2u_{xt} - 3u_{tt} = \cancel{u_{\xi\xi}} + 2\cancel{u_{\xi\eta}} + \cancel{u_{\eta\eta}} + 2\cancel{u_{\xi\xi}} + \frac{4}{3}\cancel{u_{\xi\eta}} - \frac{2}{3}\cancel{u_{\eta\eta}} - 3\cancel{u_{\xi\xi}} + 2\cancel{u_{\xi\eta}} - \frac{1}{3}\cancel{u_{\eta\eta}} = \frac{16}{3}u_{\xi\eta} = 0$$

$$\Rightarrow u_{\xi\eta} = 0 \Rightarrow u(\xi, \eta) = \phi(\xi) + \psi(\eta) \Rightarrow u(x, t) = \phi(x+t) + \psi(x-t/3)$$

$$u_t(x, t) = \phi'(x+t) - \frac{1}{3}\psi'(x-t/3)$$

$$\begin{cases} u(x, 0) = \phi(x) + \psi(x) = \sin x \\ u_t(x, 0) = \phi'(x) - \frac{1}{3}\psi'(x) = 1 \Rightarrow \phi(x) - \frac{1}{3}\psi(x) = x + K \end{cases}$$

$$\Rightarrow \psi(x) + \frac{1}{3}\psi(x) = \sin x - x - K \Rightarrow \psi(x) = \frac{3}{4}(\sin x - x - K)$$

$$\phi(x) = \sin x - \psi(x) = \sin x - \frac{3}{4}\sin x + \frac{3}{4}(x + K) = \frac{1}{4}(\sin x + 3x + 3K)$$

$$u(x, t) = \frac{1}{4}\sin(x+t) + \frac{3}{4}(x+t) + \cancel{\frac{3K}{4}} + \frac{3}{4}\sin(x-t/3) - \frac{3}{4}(x-t/3) - \cancel{\frac{3K}{4}} = \frac{1}{4}\sin(x+t) + \frac{3}{4}x + \frac{3}{4}t$$

$$+ \frac{3}{4}\sin(x-t/3) - \frac{3}{4}x + \frac{t}{4} \Rightarrow u(x, t) = \frac{1}{4}\sin(x+t) + \frac{3}{4}\sin(x-t/3) + t, \text{ no gabarito, tem um } -\frac{x}{2} \text{ que não sei se está correto.}$$